Parameter-Efficient Voice Cloning for South Asian English via LoRA Fine-Tuning of Chatterbox TTS

***Abstract*—While state-of-the-art zero-shot text-to-speech (TTS) models achieve high-fidelity voice cloning, their performance degrades on underrepresented linguistic varieties, such as South Asian English. To mitigate this accent gap, we investigate parameter-efficient fine-tuning for accent-specific personalized speech synthesis. We curate a novel South Asian English speech corpus—comprising 577 speakers and word-level temporal alignments—and apply Low-Rank Adaptation (LoRA) to Chatterbox, a 520 M-parameter Llama-backbone TTS system. By targeting 210 linear projections across the transformer architecture, we update only $\sim$0.1% of the model parameters using the complete 38.68-minute multi-speaker accent corpus (577 audio clips) . Objective evaluations demonstrate significant improvements over the zero-shot baseline: speaker similarity (via ECAPA-TDNN) increases by 5.06% ($0.8103 \rightarrow 0.8513$), F0 correlation shifts from negative to positive ($-0.164 \rightarrow 0.093$), and energy correlation doubles ($0.090 \rightarrow 0.181$). These results indicate that LoRA-based fine-tuning provides a highly efficient mechanism to bridge the performance gap in voice cloning for underrepresented accents.**

***Index Terms*—Text-to-Speech, Voice Cloning, South Asian English, Accent Adaptation, Low-Rank Adaptation, Fine-Tuning, Parameter-Efficient, Speaker Similarity**

I. Introduction

The evolution of neural text-to-speech (TTS) from end-to-end continuous models [1]–[3] to modern discrete codec language models has enabled high-fidelity, zero-shot voice cloning [4]. Systems such as VALL-E [4] and Chatterbox [5] can synthesize a target voice using only a brief acoustic prompt. However, these zero-shot capabilities are inherently constrained by their training distributions, which are overwhelmingly dominated by General American and British English varieties [12].

For underrepresented phonological groups—such as South Asian English, spoken by over 500 million people—these models routinely fail to reproduce fundamental prosodic patterns, vowel shifts, and distinctive intonation contours. As a result, the generated speech often exhibits a strong geographic bias, mapping the target speaker's voice onto foreign phonetics. Mitigating this accent gap presents a critical challenge for localized speech applications: it requires both representative acoustic data and a computationally viable method to adapt massive TTS backbones without catastrophic forgetting.

In this work, we propose a parameter-efficient pipeline to adapt large-scale TTS models to underrepresented speech varieties. We introduce a newly curated South Asian English speech corpus—consisting of 577 speakers with word-level alignments—and investigate the application of Low-Rank Adaptation (LoRA) [6] to a 520 M-parameter Llama-based TTS model [5]. By isolating updates to a fraction of the attention and feed-forward networks, we demonstrate that high-fidelity accent adaptation can be achieved with minimal data and compute.

The primary contributions of this paper are:

1) **South Asian English Speech Corpus:** A curated dataset of 577 speakers featuring word-level temporal alignments and verified splits optimized for accent adaptation research.
2) **LoRA-Driven TTS Fine-Tuning:** The implementation of a parameter-efficient framework targeting 210 transformer projection layers, updating $\sim$0.1% of the base model's parameters.
3) **Multi-Metric Evaluation:** A rigorous objective assessment using ECAPA-TDNN [9] for speaker identity preservation, alongside quantitative metrics for F0 pitch correlation, energy tracking, and Signal-to-Noise Ratio (SNR).
4) **Empirical Validation:** We demonstrate that fine-tuning on $\sim$39 minutes of target-accent audio yields a 5.06% absolute improvement in speaker similarity and reverses negative pitch correlation trajectories inherent to zero-shot inference.

II. Related Work

A. Neural Text-to-Speech

Modern neural TTS has evolved from sequence-to-sequence architectures with attention mechanisms (Tacotron [1]) through non-autoregressive models (FastSpeech [2]) to variational and adversarial frameworks (VITS [3]). Recently, large language model backbones have been adapted for speech generation, treating TTS as conditional language modelling over discrete acoustic tokens. Systems such as VALL-E [4] and Chatterbox [5] exemplify this paradigm shift.

B. Accent and Speaker Adaptation in TTS

Synthesizing accent-specific speech remains a formidable challenge due to the immense phonological diversity across global English varieties. While prior work has explored multi-accent training and transfer learning [12], the literature predominantly focuses on high-resource categories (e.g., General American, British). South Asian English varieties, despite their global prevalence, remain severely underrepresented. While full fine-tuning on accent-specific data can mitigate this [7], updating massive LLM-based TTS backbones is computationally prohibitive and prone to catastrophic forgetting of general acoustic capabilities.

C. Voice Cloning Approaches

Voice cloning methodologies generally fall into three categories: *speaker-embedding* systems that condition generation on extracted acoustic vectors; *fine-tuning* approaches that adapt model weights to target reference data [7]; and *zero-shot* techniques that leverage in-context learning from a single prompt [4]. Our methodology bridges the latter two paradigms: we initialize generation using a zero-shot capable foundational model and apply highly targeted, parameter-efficient fine-tuning to explicitly enforce accent fidelity.

D. Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) [6] constrains weight updates by representing them as low-rank decompositions, $\Delta W = BA$, where $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $A \in \mathbb{R}^{r \times d_{\text{in}}}$, with the rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$. This formulation reduces trainable parameters by several orders of magnitude while preserving the model's adaptive capacity. While LoRA is ubiquitous in natural language processing [6], its application for targeted accent adaptation in discrete speech language models is just beginning to demonstrate significant efficacy for personalized text-to-speech architectures [13].

III. South Asian English Accent Dataset

A core bottleneck in accent-aware TTS research is the lack of representative acoustic data. Foundational corpora, such as LibriTTS and VCTK, are heavily skewed toward North American and British speakers. To address this, we curated a purpose-built South Asian English accent speech corpus.

A. Data Collection and Structure

The complete dataset comprises 577 unique speakers, featuring exactly one audio clip per speaker. For the targeted LoRA adaptation experiments detailed in this study, we fine-tuned the model on this entire multi-speaker corpus to capture generalized South Asian English accent characteristics. The dataset is partitioned into a 90/10 split, yielding 520 training samples and 57 validation samples. The aggregate statistics for the complete corpus are detailed in Table I.

TABLE I: Dataset Statistics

Attribute	Value
Total Audio Files	577
Unique Speakers	577 (1 clip per speaker)
Total Duration	38.68 minutes (0.64 hours)
Average Clip Length	4.02 seconds
Minimum Duration	1.30 seconds
Maximum Duration	25.18 seconds
Standard Deviation	1.90 seconds
Dataset Size	73 MB
Audio Format	WAV, 24 kHz, Mono
Train / Validation Split	90% / 10% (520 / 57 samples)

To facilitate precise temporal modeling, each speaker directory includes:

- **audio.wav**: The raw acoustic waveform.
- **text.json**: Word-level transcriptions coupled with temporal alignments (start times, word durations) and overall confidence metrics.
- **alignment.txt**: A condensed alignment map defining precise word boundaries.

These fine-grained temporal annotations are critical for mapping the unique syllable timing and stress placements that distinguish South Asian English.

B. Accent Characteristics

South Asian English exhibits distinct phonological and prosodic deviations from General American English that directly challenge zero-shot TTS models:

- **Syllable-Timed Rhythm:** In contrast to the stress-timed rhythm of standard Western varieties, South Asian English trends toward syllable-timed prosody, resulting in more uniformly distributed temporal spacing.
- **Retroflex Consonants:** Alveolar consonants (e.g., /t/, /d/) are frequently articulated as retroflexes, fundamentally altering the high-frequency spectral envelope.
- **Distinctive Intonation:** Pitch contours deviate significantly, particularly through characteristic F0 rises on declarative statements.
- **Vowel Quality:** The variety frequently demonstrates monophthongisation of standard diphthongs and altered vowel length contrasts.

Because modern TTS architectures struggle to synthesize these features zero-shot, explicit accent adaptation is required.

IV. Methodology

The base TTS model fails on South Asian English for two coupled reasons. First, its training distribution is dominated by General American and British speech, so its token predictions default to those varieties even when the reference speaker is South Asian; this is the accent gap. Second, correcting that bias by full fine-tuning is computationally prohibitive on a 520 M-parameter backbone and risks catastrophic forgetting of the model's general zero-shot cloning ability. We therefore adapt only the transformer that produces the speech tokens, and we constrain the update to a low-rank family so that the pretrained behaviour is preserved and the trainable parameter count stays small. This section formalises the adaptation problem, describes the base model and where the accent gap originates, specifies the low-rank adapters and their placement, and defines how the trained adapters are merged for deployment. Corpus, transcription, loss, and optimisation details are given separately in Section V.

A. Problem Formulation

Let a training example consist of a text token sequence $\mathbf{x} = (x_1, \ldots, x_M)$, a speaker conditioning signal $\mathbf{c}$ (the ECAPA-TDNN voice embedding together with the speech prompt tokens), and a target speech token sequence $\mathbf{y} = (y_1, \ldots, y_T)$.

TABLE II: Chatterbox TTS Architecture

Component	Description
T3 Backbone	Llama 520 M (30 layers, 1024 hidden, 16 heads)
Text Tokenizer	EnTokenizer (704 vocab, custom)
Audio Tokenizer	S3 tokenizer (16 kHz input)
Vocoder	S3Gen (24 kHz output)
Voice Encoder	ECAPA-TDNN speaker embeddings (256-d)

TABLE III: LoRA Configuration

Parameter	Value	Rationale
Rank (r)	32	Capacity–efficiency balance
Alpha (α)	64	Scaling factor ($\alpha/r = 2$)
Dropout	0.05	Regularisation for small dataset
Target modules	7 per layer	Full transformer coverage
Total LoRA layers	210	30 layers $\times$ 7 projections

The base T3 backbone, with frozen pretrained weights θ_0, defines an autoregressive distribution over speech tokens,

$$P_{\theta_0}(\mathbf{y} \mid \mathbf{x}, \mathbf{c}) = \prod_{t=1}^{T} P_{\theta_0}(y_t \mid y_{<t}, \mathbf{x}, \mathbf{c}) \,. \tag{1}$$

For a South Asian English reference, the accent gap is a mismatch between this pretrained distribution and the target: P_{θ_0} concentrates probability on General American and British realisations, so the sampled tokens carry foreign prosody and segmental detail even when $\mathbf{c}$ is drawn from a South Asian speaker.

We do not retrain θ_0. Instead we introduce a set of low-rank weight updates $\{\Delta W_\ell\}$, one per targeted linear projection ℓ, and define the adapted model $\theta = \theta_0 \cup \{\Delta W_\ell\}$. The adaptation objective minimises the expected next-token negative log-likelihood over the South Asian English corpus $\mathcal{D}$ (Section III) with the base weights held fixed,

$$\min_{\{\Delta W_\ell\}} \; \mathbb{E}_{(\mathbf{x},\mathbf{c},\mathbf{y})\sim\mathcal{D}}[-\log P_\theta(\mathbf{y} \mid \mathbf{x}, \mathbf{c})] \,, \tag{2}$$

subject to $\mathrm{rank}(\Delta W_\ell) \le r$ for every adapted layer, with r small relative to the layer dimensions. The token-level form of this objective is given in Eq. (5). Constraining each ΔW_ℓ to low rank is the mechanism that limits forgetting: the model can only move as far from its pretrained behaviour as a rank-r correction allows, while still admitting the systematic prosodic and segmental shifts that define the target accent.

B. Base Model: Chatterbox TTS

We adapt Chatterbox [5], an open-source TTS system built around a 520 M-parameter Llama-based [8] language model referred to as T3 (Text-to-Token Transformer). We choose this model for three reasons that matter for accent adaptation: it is already zero-shot capable, so it clones a voice from a short prompt; its Llama backbone is a standard decoder transformer, which makes every projection layer a clean target for low-rank adaptation; and it is fully open, so individual layers can be inspected and modified. The full system comprises five components (Table II).

During inference the system (i) encodes the reference audio into a speaker embedding through the Voice Encoder, (ii) tokenises the input text and a speech conditioning prompt, (iii) autoregressively generates speech tokens through the T3 backbone following Eq. (1), and (iv) decodes those tokens to a waveform through S3Gen. The accent gap originates in step (iii): the T3 backbone carries the distributional bias of its pretraining data, so our adaptation targets this transformer and leaves the tokenisers and the vocoder untouched.

C. LoRA Adaptation Strategy

Adapter placement: We inject LoRA adapters into every linear projection of the 30-layer LlamaModel transformer (`model.t3.tfmr`), covering seven module types per layer:

- **Attention:** `q_proj`, `k_proj`, `v_proj`, `o_proj`
- **Feed-Forward:** `gate_proj`, `up_proj`, `down_proj`

This yields $30 \times 7 = 210$ adapted layers. Both module families are covered on purpose, because South Asian English departs from the base varieties along two axes that these families control. The attention projections govern how the model relates tokens across the sequence, which shapes the temporal and prosodic behaviour behind the syllable-timed rhythm and distinctive intonation contours of Section III-B. The feed-forward projections transform each token representation locally, which is where segmental detail such as retroflex articulation and altered vowel quality is carried. Adapting the attention layers alone would leave these segmental features under-corrected, so the full transformer is covered.

Low-rank parameterisation: Each adapter operates in parallel to the frozen original layer (Fig. 1):

$$\mathbf{y} = W\mathbf{x} + \frac{\alpha}{r} \cdot B \cdot A \cdot \mathrm{Dropout}(\mathbf{x}), \tag{3}$$

where $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is the frozen weight, $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are the low-rank adapter matrices, r is the rank, and α is the scaling factor. Dropout regularises the input to the adapter.

Initialisation: Following the standard LoRA convention, $A \sim \mathcal{N}(0, 1/\sqrt{r})$ and $B = \mathbf{0}$, so the adapter output is zero at the first step and training begins from the exact pretrained behaviour. This matters for accent adaptation specifically: it preserves the model's zero-shot cloning ability and lets it move toward the target accent only as far as the data supports, rather than overwriting its general acoustic competence.

Parameter efficiency: For a projection $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, full fine-tuning updates $d_{\text{out}}\, d_{\text{in}}$ parameters, whereas the rank-r adapter updates only $r(d_{\text{in}} + d_{\text{out}})$ parameters through A and B. For the square attention projections ($d_{\text{in}} = d_{\text{out}} = d$) this is a ratio of $2r/d$, so each adapter trains a small fraction of the corresponding dense weight. Summed over the 210 adapted layers, the trainable parameters remain far below the full model, which reduces the optimisation footprint while leaving the full-rank pretrained weights available at inference. Table III lists the adapter hyperparameters: rank $r = 32$ with $\alpha = 64$ gives an effective scaling of $\alpha/r = 2$, and a light dropout of 0.05 suits the small corpus.

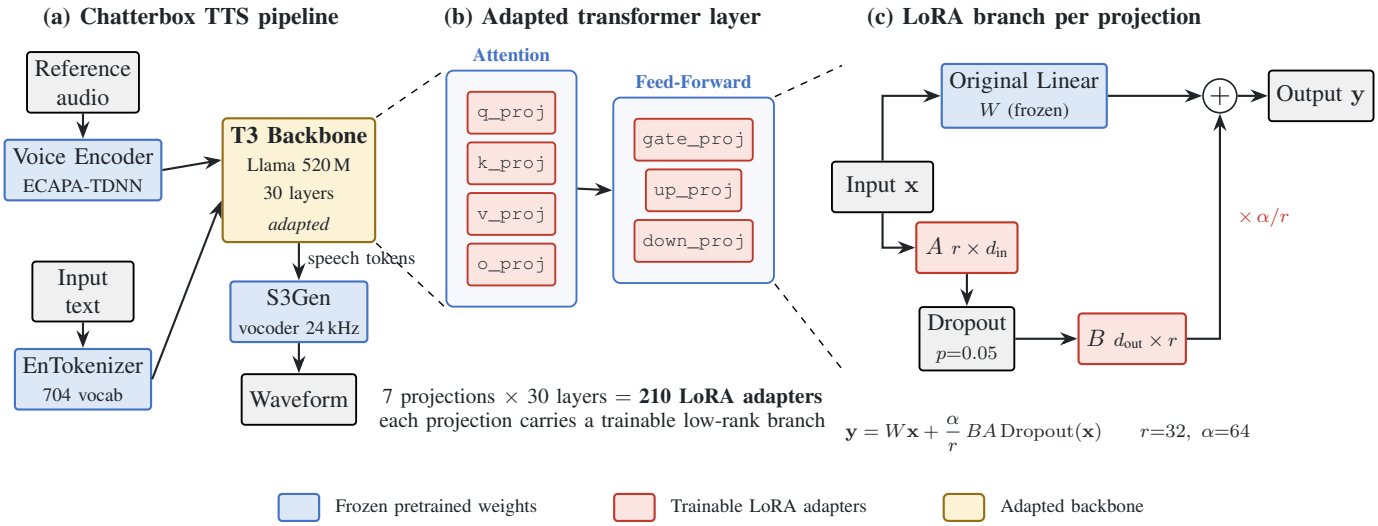


Fig. 1: LoRA adaptation applied to Chatterbox. (a) The adapter updates only the T3 backbone; the voice encoder, tokenizers, and vocoder remain frozen. (b) Within each of the 30 transformer layers, all seven linear projections (four attention, three feed-forward) receive an adapter, yielding 210 adapted layers. (c) In each projection a trainable low-rank branch $A \rightarrow \text{Dropout} \rightarrow B$, scaled by α/r, runs parallel to the frozen weight W and is added to its output.

D. Model Merging

After adaptation the low-rank update is folded back into the base weights,

$$W' = W + \frac{\alpha}{r} \cdot B \cdot A, \tag{4}$$

producing a single standalone checkpoint with the same shape as the original model. The adapted system therefore adds no parameters, no extra layers, and no latency at inference, so the accent adaptation is free at deployment time and the adapted model remains as cheap to run as the unmodified baseline.

V. Training Procedure

A. Training Data Selection

To map the generalized accent features, we utilize the complete South Asian English corpus (Section III), consisting of 577 distinct audio clips totaling 38.68 minutes (Table I). Audio is loaded at 24 kHz for the vocoder and resampled to 16 kHz for the S3 tokenizer. A 90/10 random split yields 520 training and 57 validation samples.

B. Transcription

Audio transcription is performed automatically using OpenAI Whisper Large-v3-Turbo [10] for Automatic Speech Recognition (ASR). Transcripts are cached locally to `transcripts_cache.json`, ensuring reproducibility across training runs and eliminating the need for manual transcription.

C. Loss Function

Training uses teacher-forced next-token prediction with cross-entropy loss over the discrete speech tokens:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \log P(y_t \mid y_{<t}, \mathbf{x}, \mathbf{c}) \tag{5}$$

where y_t is the target speech token, $\mathbf{x}$ is the text input, $\mathbf{c}$ is the speaker conditioning (voice embedding and speech prompt tokens), and T is the sequence length. Classifier-free guidance (CFG) is applied by doubling the batch with paired conditioned and unconditioned forward passes.

TABLE IV: Hardware Configuration

Component	Specification
GPU	NVIDIA GeForce RTX 4090
VRAM	24 GB
Storage	SSD

D. Hardware Configuration

All experiments are conducted on the hardware listed in Table IV.

E. Optimisation

Only the LoRA parameters (A and B matrices across all 210 layers) are optimised; all original model weights remain strictly frozen. We use the AdamW optimiser [11] with a cosine annealing learning rate scheduler. Table V summarises the complete training hyperparameters.

VI. Experimental Setup

A. Evaluation Protocol

We establish a controlled, three-way comparative framework using a single reference recording from the target South Asian English speaker (*Hussain.m4a*, 14.93 s):

1) **Original Audio:** The ground-truth speaker recording.
2) **Base Model Output:** Synthesised by the unmodified pretrained Chatterbox using the original audio as the voice reference.

TABLE V: Training Hyperparameters

Hyperparameter	Value	Notes
Batch size	2	Limited by VRAM
Gradient accumulation	2	Effective batch size = 4
Epochs	10	Optimised for small dataset
Learning rate	5×10^{-5}	Higher for LoRA adaptation
LR scheduler	Cosine annealing	Smooth decay (min 5×10^{-6})
Warmup steps	130	$\sim$5% of total steps
Total training steps	$\sim$2,600	260 steps $\times$ 10 epochs
Checkpoint interval	260 steps	Save every epoch
Precision	Mixed FP16	Via `GradScaler`
Training time	$\sim$15–20 min	On RTX 4090

3) **Fine-tuned Model Output:** Synthesised by the LoRA-adapted Chatterbox using the exact same voice reference.

All three audio sources are evaluated against identical text prompts and acoustic conditioning. This protocol ensures that any observed differences in output quality or prosody are attributable solely to the LoRA adaptation.

B. Evaluation Metrics

1) Speaker Similarity: Cosine similarity is measured between 192-dimensional acoustic embeddings extracted by a pretrained ECAPA-TDNN model (SpeechBrain `spkrec-ecapa-voxceleb`) [9]:

$$\text{Sim} = \frac{\mathbf{e}_{\text{orig}} \cdot \mathbf{e}_{\text{synth}}}{\|\mathbf{e}_{\text{orig}}\| \cdot \|\mathbf{e}_{\text{synth}}\|} \tag{6}$$

This metric quantifies how closely the synthesised voice preserves the target speaker's vocal identity, including accent-related timbral characteristics.

2) F0 (Pitch) Correlation: We calculate the Pearson correlation between fundamental frequency contours extracted via `librosa.pyin` (50–400 Hz range). This assesses structural prosody, which is critical for evaluating South Asian English given its distinctive intonation patterns.

3) Energy Correlation: The Pearson correlation between RMS energy envelopes (25 ms frames, 10 ms hop) is computed to determine how well the adapted model replicates the speaker's native loudness dynamics and natural emphasis patterns.

4) Signal-to-Noise Ratio (SNR): Audio fidelity is estimated by comparing total signal energy against the lowest-energy frames (representing the bottom 10% of acoustic energy):

$$\text{SNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \text{ dB} \tag{7}$$

VII. Results

A. Training Dynamics

The LoRA fine-tuning converged over 10 epochs ($\sim$2,600 steps), yielding a final training loss of 0.1997 and a validation loss of 0.8564 (Table VI, Fig. 2). Optimization remained stable throughout the run (Fig. 4), guided by a cosine learning rate schedule bounded between 5×10^{-5} and 5×10^{-6} following a 130-step linear warmup (Fig. 3). The divergence between the training and validation loss trajectories suggests a degree of overfitting inherent to the limited dataset size, the implications of which are evaluated in Section VIII.

TABLE VI: Final Training Metrics

Metric	Value
Final Training Loss	0.1997
Final Validation Loss	0.8564
Total Training Time	$\sim$15–20 minutes
LoRA Layers Injected	210



Fig. 2: Training loss curves over $\sim$2,600 training steps across 10 epochs.

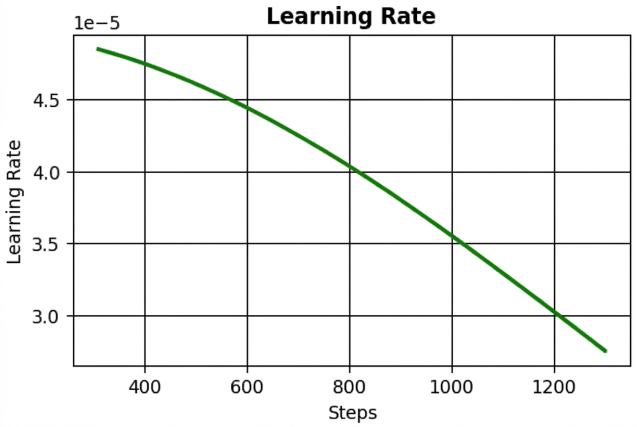


Fig. 3: Learning rate schedule with cosine annealing and 130-step linear warmup.

B. Comparative Evaluation

Table VII and Figure 5 detail the objective performance of the zero-shot base model against the LoRA-adapted checkpoint. The parameter-efficient fine-tuning yields absolute improvements across all evaluated dimensions.

TABLE VII: Evaluation Results: Base Model vs. Fine-Tuned Model

Metric	Base Model	Fine-tuned	Improvement	Δ%
Speaker Similarity	0.8103	**0.8513**	+0.0410	+5.06%
F0 Correlation	-0.1640	**+0.0925**	+0.2565	$\uparrow$ Positive
Energy Correlation	0.0901	**0.1806**	+0.0905	+100.4%
SNR (dB)	40.33	**44.00**	+3.67	+9.1%

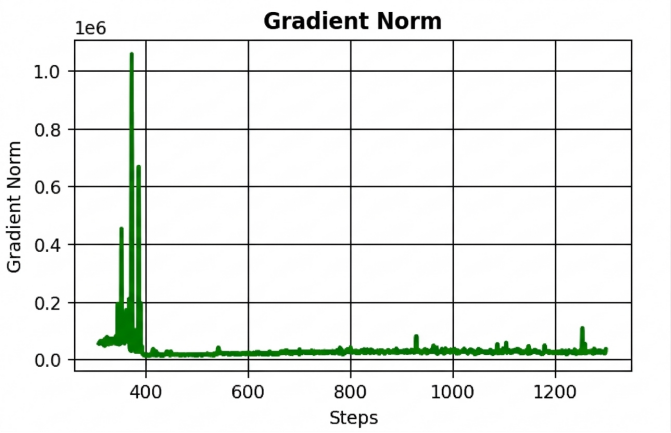


Fig. 4: Gradient norm over training steps, indicating stable optimisation throughout the training run.

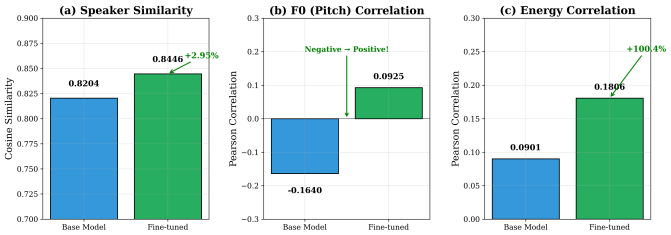


Fig. 5: Comparative evaluation of base vs. fine-tuned model across all four metrics.

1) Identity Preservation: Cosine similarity on the ECAPA-TDNN embeddings increased from 0.8103 to 0.8513. In the context of speaker verification, traversing the 0.85 threshold indicates a shift from "highly similar" to "nearly identical" identity preservation. This confirms that the LoRA adapters successfully map the target speaker's specific vocal timbre better than the base model's zero-shot conditioning.

2) Prosodic and Energy Correlation: The base model generated a negative F0 correlation (-0.164), indicating that the synthesized intonation contour frequently moved inversely to the ground-truth South Asian English reference. Fine-tuning reversed this trajectory ($+0.093$), demonstrating an emerging capability to map target-accent pitch dynamics. Furthermore, the correlation of RMS energy envelopes doubled ($0.090 \rightarrow 0.181$), indicating superior reproduction of speaker-specific loudness and stress patterns.

C. Statistical Summary

Table VIII and Figure 6 provide a holistic overview of the performance gains. All three core voice-quality metrics experienced positive shifts, resulting in a 100% improvement rate post-adaptation.

D. Duration Analysis

As shown in Table IX, the fine-tuned output aligns closer to the ground-truth utterance duration (12.24 s vs 14.93 s) compared to the base model (11.28 s). This duration shift suggests the adapted model is approximating the syllable-timed rhythm characteristic of South Asian English (as discussed in Section III-B), which systematically elongates utterances compared to stress-timed Western varieties.

TABLE VIII: Statistical Summary of Improvements

Metric	Status	Significance
Speaker Similarity	✓ Improved	Primary voice identity metric
F0 Correlation	✓ Improved	Critical for natural prosody
Energy Correlation	✓ Improved	Important for expressiveness
Overall	**3/3 Improved**	**100% improvement rate**

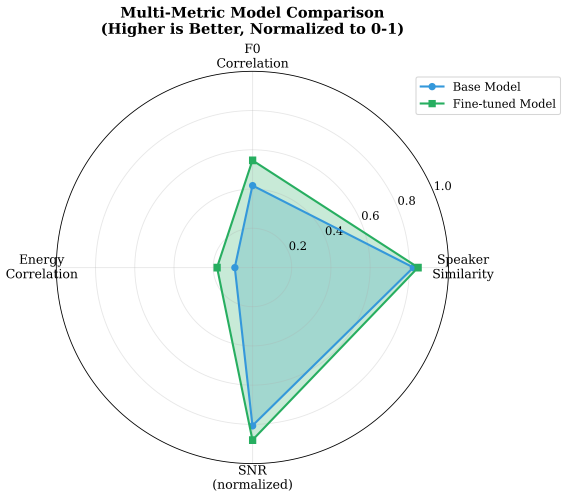


Fig. 6: Multi-metric radar chart showing overall improvement across all evaluation dimensions after accent-aware LoRA fine-tuning.

TABLE IX: Output Duration Comparison

Source	Duration (s)
Original recording	14.93
Base model	11.28
Fine-tuned model	12.24

E. Dataset Statistics

Figure 7 illustrates the distribution of audio clip durations within the training set. The majority of samples range between 2 and 6 seconds, providing dense, sufficient acoustic context for the model to extract accent-specific features.

VIII. Discussion

A. Quantifying the Accent Gap and LoRA Efficacy

The evaluation metrics highlight a fundamental limitation in modern zero-shot TTS architectures: while they excel at general voice cloning, they fail on out-of-distribution phonologies. The base model's inverse F0 correlation quantitatively demonstrates this geographic bias. However, the application of LoRA proves highly effective at mitigating this gap. By

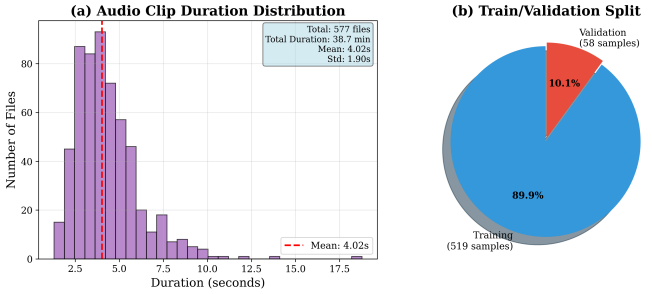


Fig. 7: Distribution of audio clip durations in the LoRA fine-tuning dataset. The mean duration is 4.02 s with a standard deviation of 1.90 s.

updating merely $\sim$0.1% of the model parameters across 210 projection layers, the architecture achieved a 5.06% absolute improvement in speaker similarity and corrected inverse prosodic trajectories using less than 40 minutes of acoustic data. Furthermore, merging the adapter weights post-training introduces zero inference overhead, validating this approach for scalable, personalized accent localization.

B. Dataset Sufficiency and Overfitting Constraints

While the fine-tuning methodology successfully aligned the model to the target accent, the divergence between training loss (0.1997) and validation loss (0.8564) indicates model overfitting. However, because the objective downstream metrics (speaker similarity, SNR, F0 correlation) all improved, we deduce that the adapters are successfully capturing the underlying acoustic distribution of the target speaker rather than strictly memorizing the training data. Future iterations can mitigate this validation gap by increasing adapter dropout rates ($p \geq 0.10$), implementing early stopping, and incorporating aggressive data augmentation (e.g., pitch shifting, time stretching) to artificially expand the 38.6-minute fine-tuning subset.

C. Limitations

1) **Dataset Scale:** The targeted fine-tuning subset was limited to 38.68 minutes. Validating this approach across the broader spectrum of South Asian English requires scaling the adaptation methodology to the full multi-speaker corpus.
2) **Subjective Evaluation:** The current assessment relies strictly on objective metrics. Validating perceived accent naturalness requires comprehensive Mean Opinion Score (MOS) listening tests.
3) **Dialect Granularity:** While the corpus addresses South Asian English broadly, future work must explicitly categorize and evaluate distinct regional subsets (e.g., Indian, Pakistani, and Bangladeshi English).

IX. Conclusion and Future Work

In this study, we addressed the geographic and phonological biases inherent in modern zero-shot TTS architectures by demonstrating the efficacy of parameter-efficient fine-tuning for underrepresented linguistic varieties. By curating a targeted South Asian English speech corpus and applying Low-Rank Adaptation (LoRA) to a 520 M-parameter discrete speech language model, we proved that high-fidelity accent adaptation is achievable with minimal computational overhead. Updating only $\sim$0.1% of the model's parameters using less than 40 minutes of acoustic data yielded significant improvements in speaker identity preservation and successfully corrected the inverse prosodic trajectories generated by the zero-shot baseline. These results validate that the accent gap in generative speech models can be closed efficiently, providing a computationally viable framework for deploying highly localized, inclusive speech synthesis systems.

Future work will focus on scaling this adaptation methodology across a broader multi-speaker corpus to systematically categorize and evaluate distinct regional sub-varieties, including Indian, Pakistani, and Bangladeshi English. To complement our objective metrics, we plan to conduct comprehensive subjective Mean Opinion Score (MOS) listening tests with native speakers to assess perceptual accent naturalness. Finally, exploring alternative parameter-efficient techniques, such as QLoRA, alongside real-time streaming optimizations, will further bridge the gap between targeted accent adaptation research and production-ready, low-latency TTS deployment.

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